RAW INPUT

Input CRISM / M³ raw hyperspectral data

STAGE 1: EXPLORATORY DATA ANALYSIS (EDA)

- Metadata Parsing
- Cube Loading
- Physical Validation
- Wavelength Assignment
- Band Statistics
- Bad Band Detection
- Output: Validated Noise-Aware Cube

STAGE 2: NOISE CHARACTERIZATION

- Stripe Noise
- Gaussian Noise
- Impulse Noise
- Dead Bands
- Thermal / Saturation Artifacts
- Output: Noise Map

STAGE 3: PHYSICS-BASED CORRECTION

- Illumination Correction
- Atmospheric Correction
- Spike Removal
- Thermal Emission Correction
- Saturation Recovery (Interpolation)
- Output: Physics-Corrected Cube

STAGE 4: HYBRID SELF-SUPERVISED ML

- **Module 1: 1D Spatial CNN**
  Stripe Residual Learning
- **Module 2: Spectral Noise2Void (DAE)**
  Encoder → Bottleneck → Decoder
- **Adaptive Blending & Composite Loss**

QUALITY ASSESSMENT (VALIDATION)

- SNR & Pearson Correlation
- SAM & Band Depth
- Ensures Spatial Restoration while Preserving Spectra

FINAL DENOISED HYPERSPECTRAL CUBE

Improved SNR + Feature Preservation